Deepak Katukuri

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Dear Hiring Team at BMW Group,

I am writing to apply for the Werkstudent position in KI-basierte Bordnetzentwicklung within your E/E architecture team. I am an MSc Data Science candidate with 2.5 years of production software engineering experience, and my recent project work is built almost entirely around the two things this role centers on: multi-agent systems for automating engineering processes, and structured analysis of data models.

My strongest match to this role is JobPilot, a 16-node LangGraph pipeline I built that reads a document (a job description, in my use case), extracts structured claims against a Chroma-backed evidence store, and blocks any output the evidence cannot support — checkpointed for crash recovery, cost-tracked per run, and held honest by 238 pytest tests. The underlying problem is close to what your team faces with PREEvision and MagicDraw data: parsing a large, structured model, checking it for completeness and consistency, and deciding where the model itself needs to be extended. I also built a corrective-RAG pipeline that grades its own retrieved evidence and refuses to answer when the underlying data genuinely does not support a claim — the same discipline your team would need when generating Bordnetz artifacts automatically. Both projects are Python-first, which matches your stated requirement directly.

I want to name two gaps directly rather than let them surface later. First, I have no hands-on experience with SysML/UML, MagicDraw, or MBSE-style modeling specifically — my data-model experience comes from relational schemas (PostgreSQL/FastAPI) rather than systems-engineering metamodels, though the underlying skill of reasoning about a structured model's completeness and consistency is the same discipline. Second, my German is currently at A2 level. I am aware your posting asks for verhandlungssicheres Deutsch, and I want to be upfront that I am not there yet rather than overstate it. I am continuing to study German and can commit to reaching a working level over the course of a 12-month Werkstudent term; in the meantime, my English is fluent and I can be fully productive on the technical work from day one.

I am based in Potsdam and open to relocating to Munich for this role. I am available to start as a Werkstudent now, part-time alongside my MSc, and would welcome the chance to discuss how my agentic-AI and data-pipeline work could support your PBN automation efforts.

Thank you for considering my application. I look forward to hearing from you.

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